

“Calculus 3”

Multi-Variable Calculus

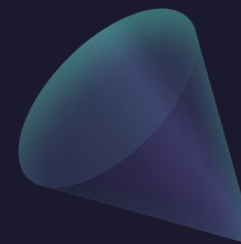
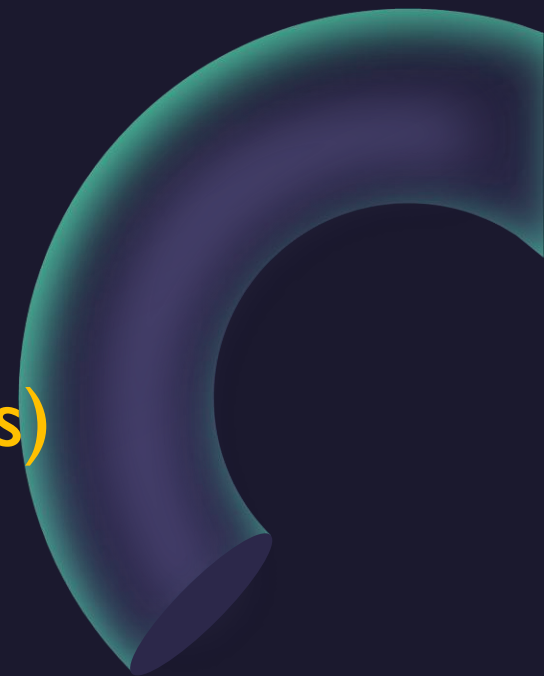
Instructor: Álvaro Lozano-Robledo

Day 2 I



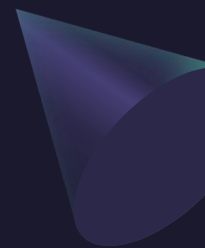
Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104





ALVARO: Start the recording!



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The Fundamental Theorem for Line Integrals



The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function F such that $F' = f$.

$$\int_a^b F'(x) \, dx = F(b) - F(a)$$

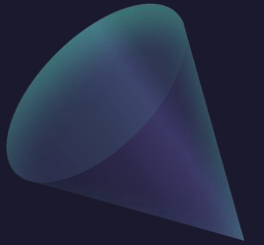
The Fundamental Theorem for Line Integrals

2 Theorem

Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

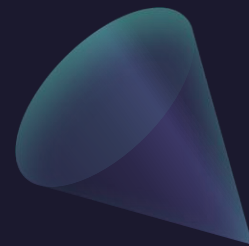
The Fundamental Theorem for Line Integrals: Proof



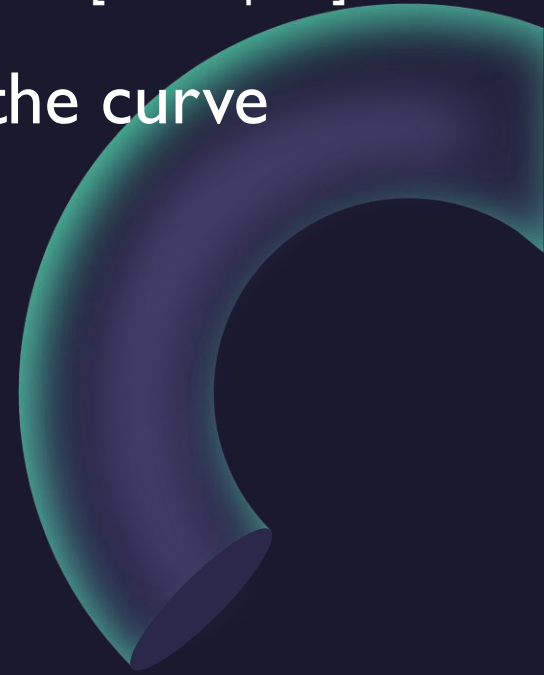
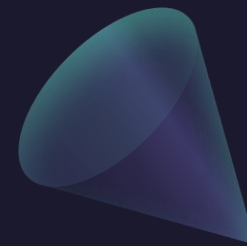
The Fundamental Theorem for Line Integrals: Proof

$$\begin{aligned}\int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\&= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\&= \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt \quad \text{(by the Chain Rule)} \\&= f(\mathbf{r}(b)) - f(\mathbf{r}(a))\end{aligned}$$

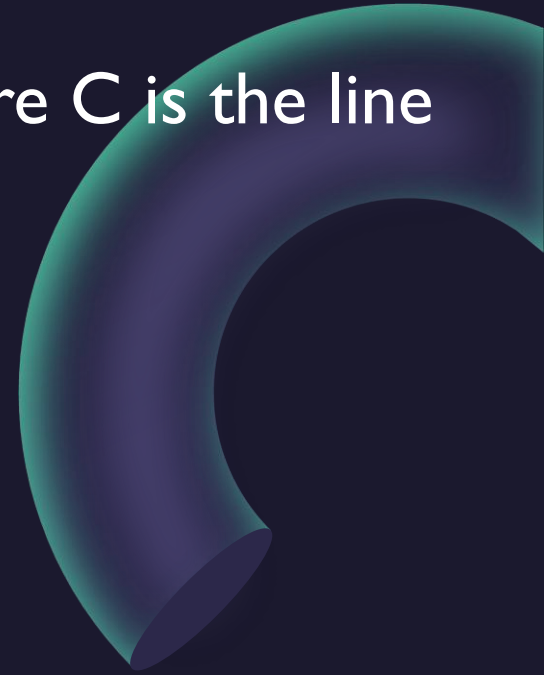
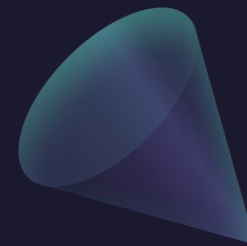
Example: Let $f(x, y) = x^2 e^y$. Evaluate $\int_C \nabla f \cdot dr$ where C is the curve parametrized by $r(t) = (2t, t^3)$, $1 \leq t \leq 2$.



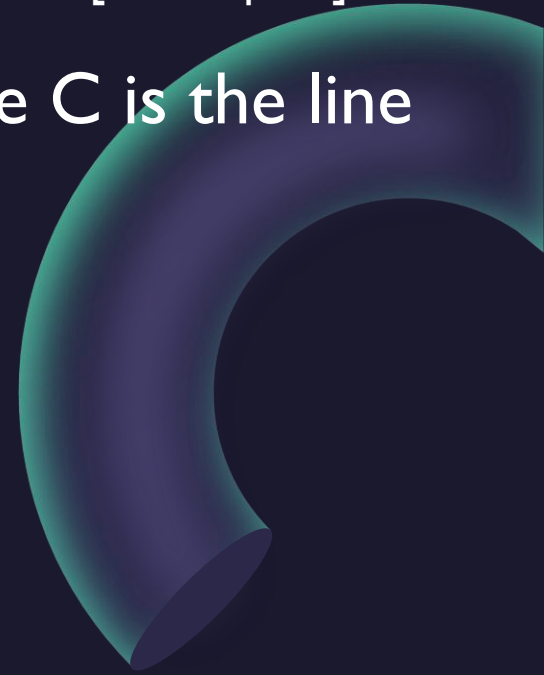
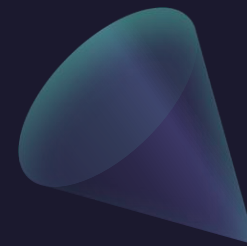
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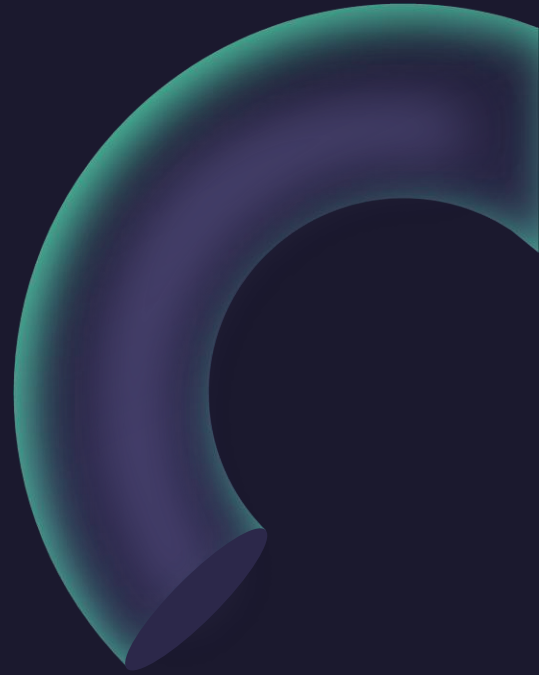
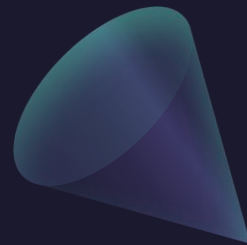
Conservative Vector Fields

A vector field $\mathbf{F}$ is called a **conservative vector field** if it is the gradient of some scalar function, that is, if there exists a function f such that $\mathbf{F} = \nabla f$. In this situation f is called a **potential function** for $\mathbf{F}$.

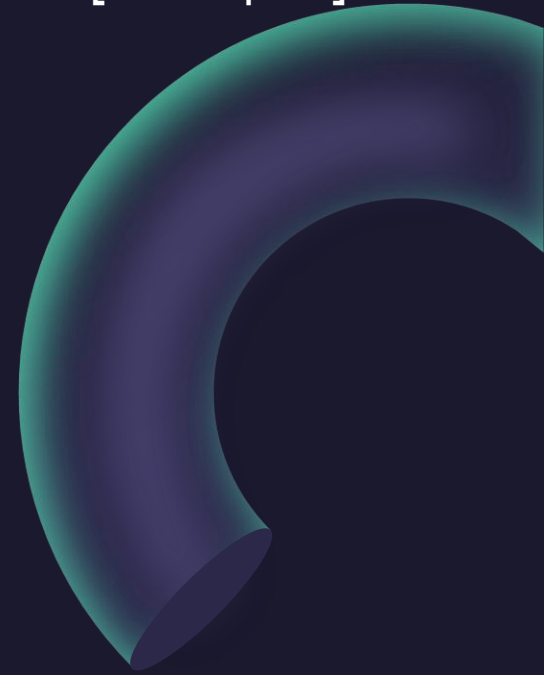
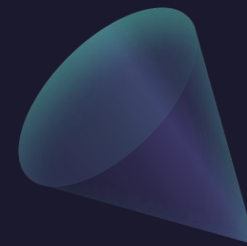
Example: Let $F(x, y, z) = (yz, xz, xy)$. Is F conservative?

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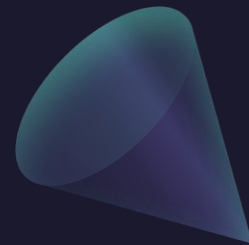
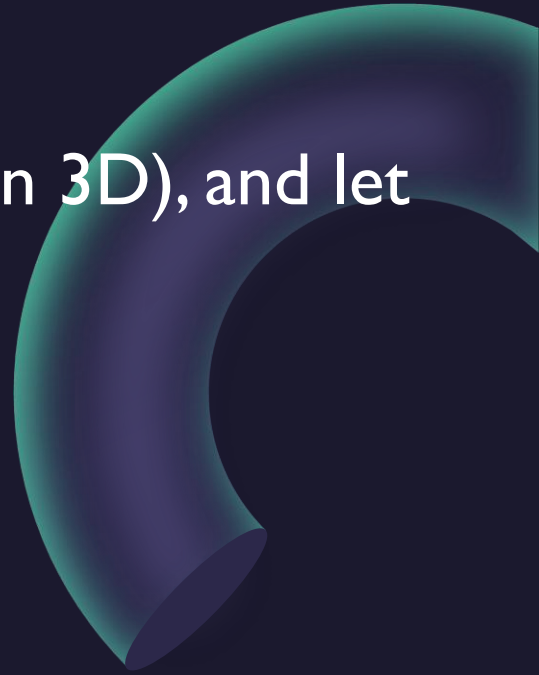


Example: Let $G(x, y, z) = (xy^2 + 3x, x^2y)$. Is G conservative?



Recall!

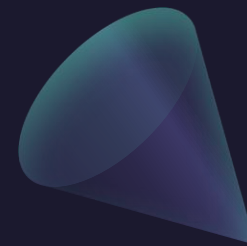
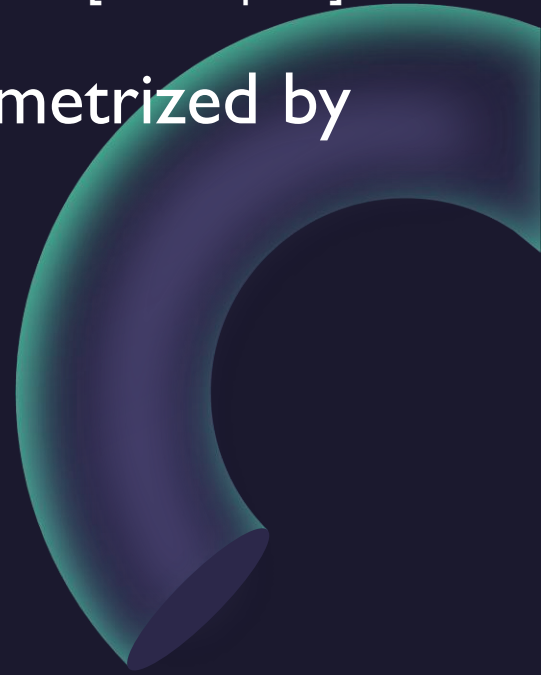
Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f$.



Example: Evaluate $\int_C 2x dx + 2y dy$ where C is the curve parametrized by $r(t) = (t^2 - 1, \sqrt{t+1})$, $0 \leq t \leq 3$.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

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$$r(t) = (t^2 - 1, \sqrt{t + 1}), \quad 0 \leq t \leq 3.$$



Path Independence

The line integrals of a vector field F are called **path-independent** if for any two paths C, D starting and ending at the same point we have

$$\int_C F \cdot dr = \int_D F \cdot dr.$$

Note: If F is conservative, then it is path-independent, by the Fundamental Theorem of Line Integrals.

Example: Let $G(x, y, z) = (xy^2 + 3x, x^2y)$. Then G is path-independent.

Path Independence

The line integrals of a vector field F are called **path-independent** if for any two paths C_1, C_2 starting and ending at the same point we have

$$\int_{C_1} F \cdot dr = \int_{C_2} F \cdot dr.$$

A parametrized curve is called **closed or a loop** if it starts and ends at the same point.

Example: Let $r(t) = (\cos(t), \sin(t))$, $0 \leq t \leq 2\pi$.
Then $r(t)$ is a loop.

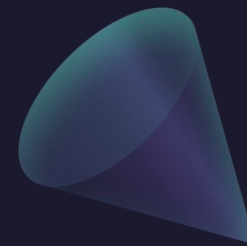
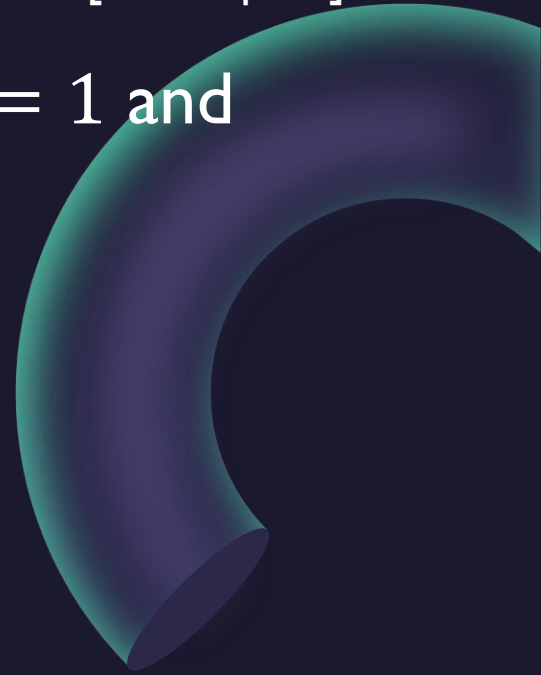
3 Theorem

$\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in D .

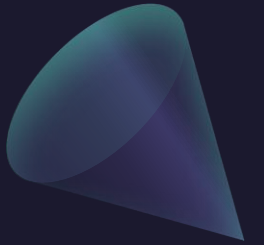
Example: Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the ellipse $4x^2 + 9y^2 = 1$ and $\mathbf{F} = ((1 + xy)e^{xy}, x^2e^{xy})$.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

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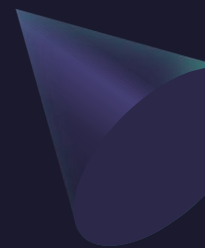


Questions?





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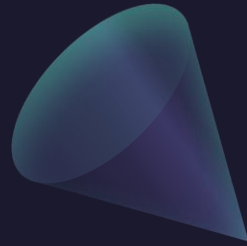
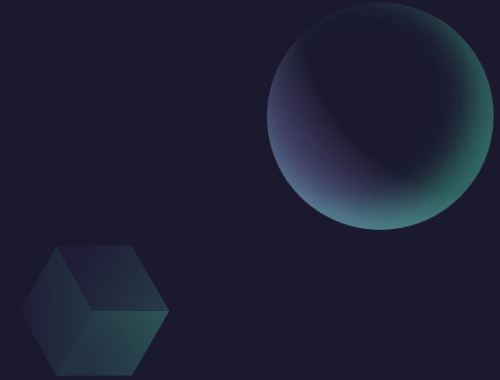


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Green's Theorem

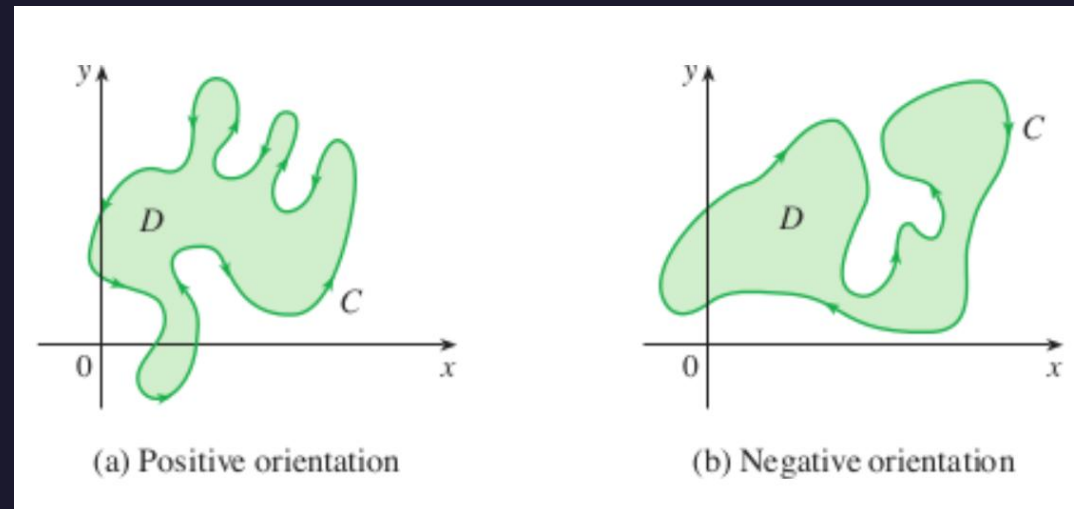


About Parametrized Curves

A parametrized curve is called **closed or a loop** if it starts and ends at the same point.

A closed curve is **simple** if its intersection with itself is at the start and the end, and nowhere else.

A simple closed curve on the plane is **positively oriented** if it goes counterclockwise. Otherwise it is **negatively oriented**.



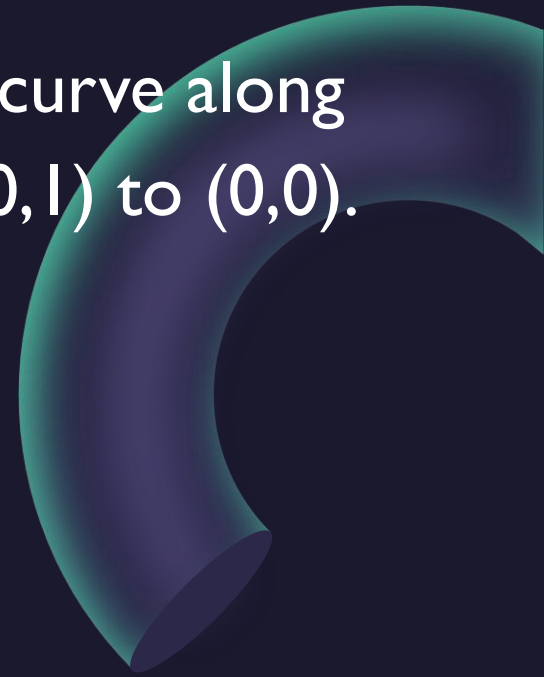
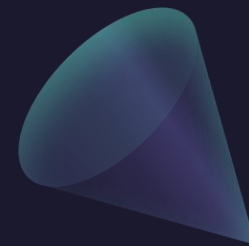
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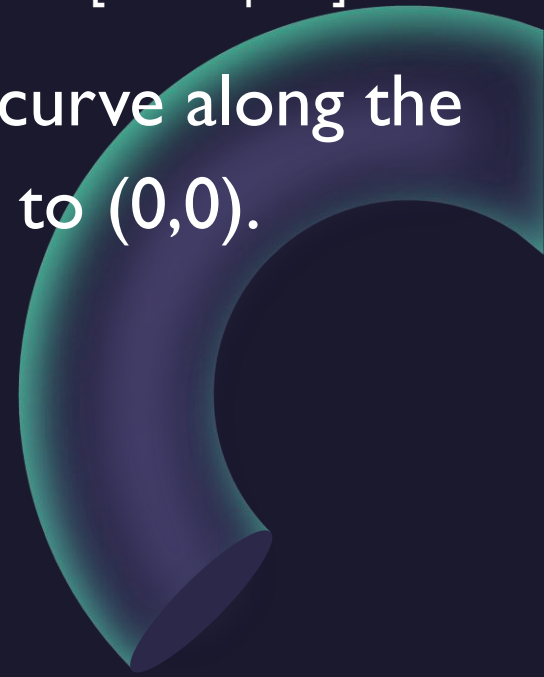
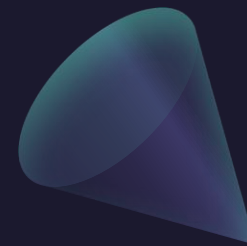
Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Example: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve along the segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.



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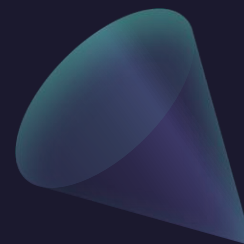
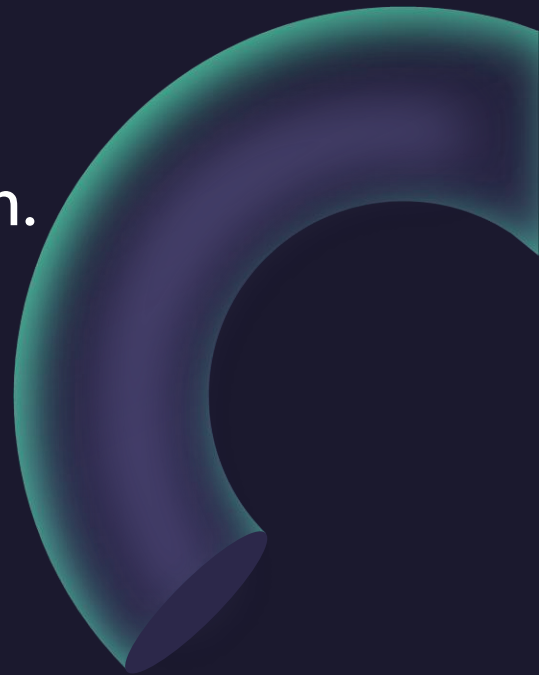
$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Notes: Sometimes we write $\oint_C \dots$ to denote we are integrating along a closed curve.

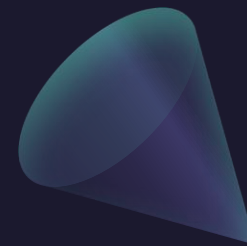
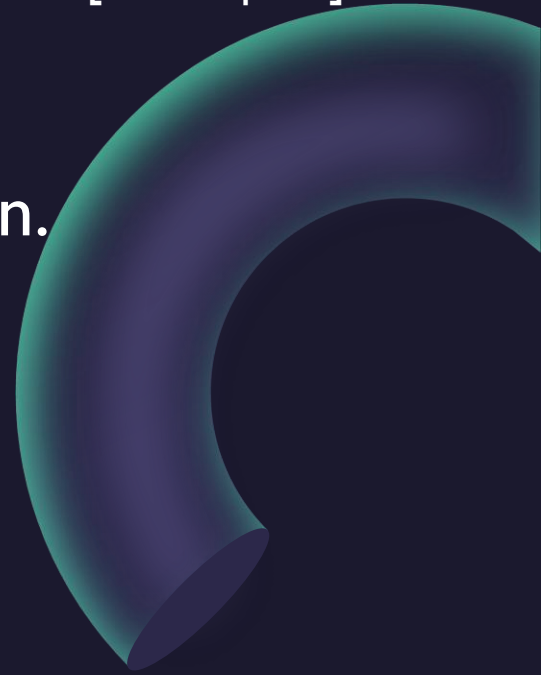
If $F = (P, Q)$ is conservative, then $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$ and therefore

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$$

Example: Evaluate $\oint_C (3y - e^{\sin(x)})dx + (7x + \sqrt{y^4 + 1})dy$
where C is the circle of radius 3 centered at the origin.



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Green's Theorem

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

Green's Theorem

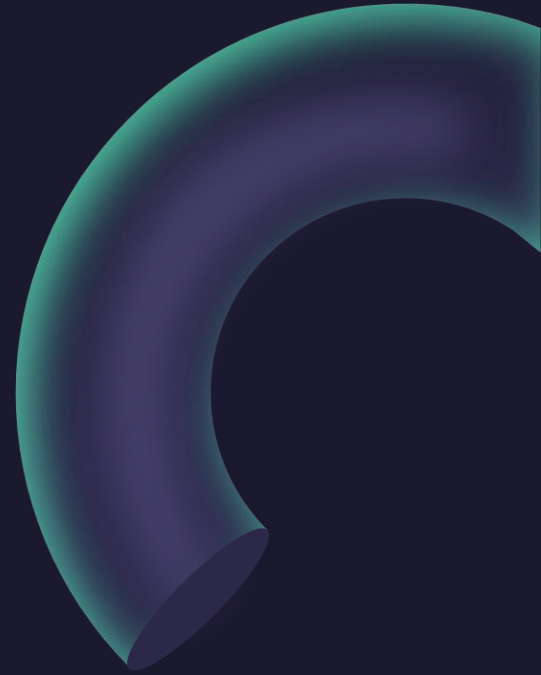
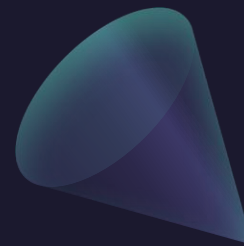
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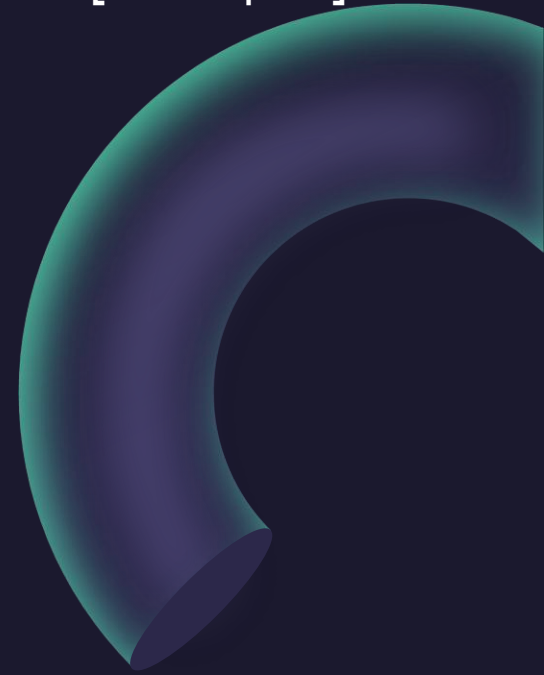
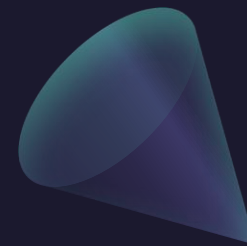
$$\begin{array}{lll} P(x, y) = 0 & P(x, y) = -y & P(x, y) = -\frac{1}{2}y \\ Q(x, y) = x & Q(x, y) = 0 & Q(x, y) = \frac{1}{2}x \end{array}$$

$$A = \oint_C x \, dy = -\oint_C y \, dx = \frac{1}{2} \oint_C x \, dy - y \, dx$$

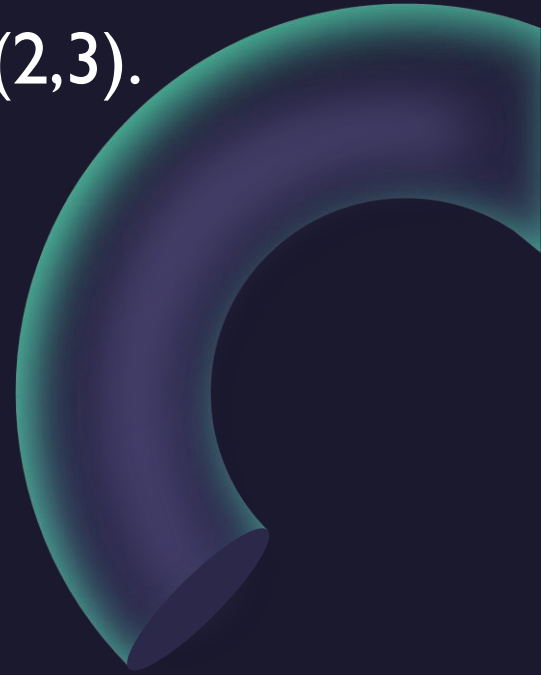
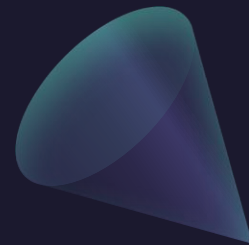
Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



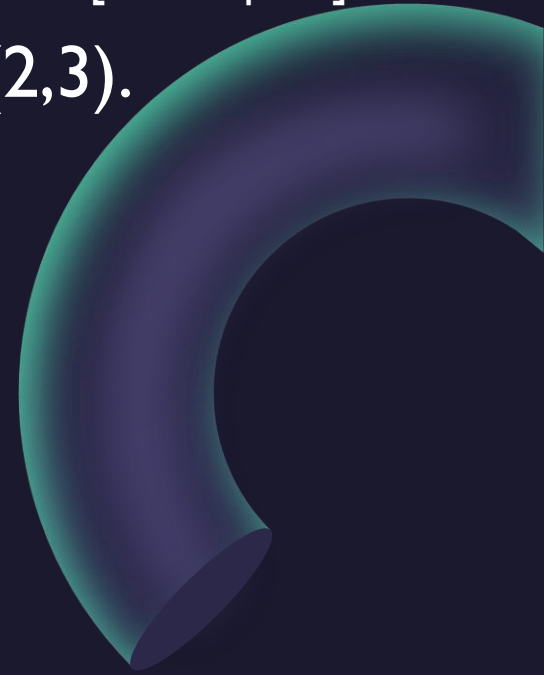
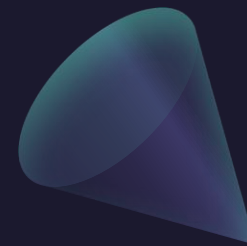
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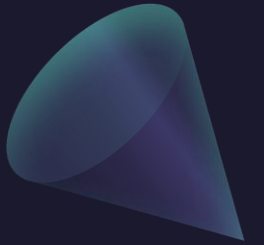
Example: Find the area of the triangle with vertices $(0,0)$, $(2,0)$, $(2,3)$.



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Questions?



Thank you

Until next time.

